

MAT11-3

Linear Algebra I

Matrices, vector spaces, and linear maps

Linear models, recognised across many objects and computed with matrices.

Albert School — 38 hours

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1 Proof literacy

This course requires a minimum of reasoning tools, used throughout.

- **Direct verification.** To prove a statement of the form “for all x , $P(x)$ ”, take an arbitrary x and establish $P(x)$ using only the definitions. The subspace test below is verified this way.
- **Counterexample.** To disprove a claim “for all x , $P(x)$ ”, exhibit a single x for which $P(x)$ fails.
- **Reading an if-then proof.** A statement “if P then Q ” is proved by assuming P and deriving Q . Its contrapositive “if not Q then not P ” is logically equivalent (see MAT11-1).

These tools are assumed in every proof that follows.

2 Matrix arithmetic

A **matrix** of size $m \times n$ is a rectangular array of real numbers with m rows and n columns, written $A = (a_{ij})$ where a_{ij} is the entry in row i , column j . The set of all such matrices is denoted $M_{m,n}(\mathbb{R})$, and $M_n(\mathbb{R})$ when $m = n$ (square matrices).

Definition 1 Let $A = (a_{ij})$ and $B = (b_{ij})$ be matrices.

- **Addition** (same size): $(A + B)_{ij} = a_{ij} + b_{ij}$.
- **Scalar multiplication:** $(\lambda A)_{ij} = \lambda a_{ij}$ for $\lambda \in \mathbb{R}$.
- **Transpose:** A^T is the $n \times m$ matrix with $(A^T)_{ij} = a_{ji}$.
- **Matrix multiplication** (if A is $m \times n$ and B is $n \times p$): AB is the $m \times p$ matrix with

$$(AB)_{ij} = \sum_{k=1}^n a_{ik}b_{kj}.$$

The number of columns of A must equal the number of rows of B for AB to be defined.

Proposition 2 Addition is commutative and associative; scalar multiplication distributes over addition. Matrix multiplication is associative, $(AB)C = A(BC)$, and distributes over addition, but is **not commutative**: in general $AB \neq BA$. Also $(AB)^T = B^T A^T$.

Worked example. Take $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. Then $AB = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ while $BA = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$, so $AB \neq BA$. This single counterexample establishes non-commutativity.

The $n \times n$ **identity matrix** I_n has 1 on the diagonal and 0 elsewhere; it satisfies $AI_n = A$ and $I_n A = A$ whenever the product is defined.

3 Linear systems

A system of m linear equations in n unknowns can be written in **matrix form**

$$AX = B,$$

where $A \in M_{m,n}(\mathbb{R})$ is the matrix of coefficients, X is the column vector of unknowns, and B is the column vector of right-hand sides.

3.1 Gaussian elimination

Gaussian elimination transforms the system into an equivalent one (same solution set) using the **elementary row operations**:

1. swap two rows;
2. multiply a row by a non-zero scalar;
3. add a multiple of one row to another.

Applied to the augmented matrix $(A \mid B)$, these operations reduce A to **row echelon form**: in each non-zero row the first non-zero entry (the **pivot**) lies strictly to the right of the pivot in the row above, and rows of zeros are at the bottom.

3.2 Pivot analysis and solution classification

A column containing a pivot corresponds to a **pivot variable**; a column without a pivot corresponds to a **free variable**. After reduction:

- if a row reads $0 = c$ with $c \neq 0$, the system has **no solution**;
- otherwise, if every variable is a pivot variable, the solution is **unique**;
- otherwise there are free variables and the system has **infinitely many solutions**, one for each choice of the free variables.

3.3 Parametric form

When free variables exist, express each pivot variable in terms of the free ones; the free variables are the parameters. For example, a solution set may be written

$$X = X_0 + t_1 V_1 + \dots + t_k V_k, \quad t_1, \dots, t_k \in \mathbb{R},$$

where X_0 is one particular solution and $V_1, \dots, V_k$ describe the free directions.

4 A matrix acting on a vector

A matrix $A \in M_{m,n}(\mathbb{R})$ defines a map $\mathbb{R}^n \rightarrow \mathbb{R}^m$ sending the column vector X to AX . This map is **linear**:

$$A(X + Y) = AX + AY, \quad A(\lambda X) = \lambda(AX).$$

Definition 3 For $A \in M_{m,n}(\mathbb{R})$:

- the **null space** (kernel) is $\ker A = \{X \in \mathbb{R}^n : AX = 0\}$;
- the **column space** (image) is the set of all AX as X ranges over $\mathbb{R}^n$, equivalently the set of linear combinations of the columns of A .

The null space is exactly the solution set of the homogeneous system $AX = 0$.

5 Vector spaces

The properties of $\mathbb{R}^n$ under addition and scalar multiplication are taken as the defining **axioms** of a vector space: $\mathbb{R}^n$ with the operations already in use satisfies them.

Definition 4 A **vector space** over $\mathbb{R}$ is a set E with an addition $E \times E \rightarrow E$ and a scalar multiplication $\mathbb{R} \times E \rightarrow E$ such that, for all $u, v, w \in E$ and $\lambda, \mu \in \mathbb{R}$:

- addition is associative and commutative;
- there is a zero vector 0 with $u + 0 = u$;
- every u has an opposite $-u$ with $u + (-u) = 0$;
- $\lambda(u + v) = \lambda u + \lambda v$ and $(\lambda + \mu)u = \lambda u + \mu u$;
- $(\lambda\mu)u = \lambda(\mu u)$ and $1 \cdot u = u$.

Elements of E are called **vectors**.

Worked example. Standard examples:

- $\mathbb{R}^n$, the space of numerical vectors;
- $\mathbb{R}[X]$, the space of polynomials, and $\mathbb{R}_n[X]$, polynomials of degree at most n ;
- function spaces, e.g. the set of functions $\mathbb{R} \rightarrow \mathbb{R}$ with pointwise addition and scalar multiplication;
- $M_{m,n}(\mathbb{R})$, the space of matrices of a fixed size.

6 Subspaces

Definition 5 A subset F of a vector space E is a **subspace** if it is itself a vector space for the operations of E . Equivalently (three-criterion test), F is a subspace if and only if:

1. $0 \in F$;
2. F is closed under addition: $u, v \in F \Rightarrow u + v \in F$;
3. F is closed under scalar multiplication: $\lambda \in \mathbb{R}, u \in F \Rightarrow \lambda u \in F$.

Criteria 2 and 3 are often combined: F is a subspace if $0 \in F$ and, for all $\lambda \in \mathbb{R}$ and $u, v \in F$, $\lambda u + v \in F$.

6.1 Span

Definition 6 The **span** (generated subspace) of vectors $v_1, \dots, v_k$ is the set of all their linear combinations,

$$\text{span}(v_1, \dots, v_k) = \{\lambda_1 v_1 + \dots + \lambda_k v_k : \lambda_i \in \mathbb{R}\}.$$

It is the smallest subspace containing $v_1, \dots, v_k$.

6.2 Column space and null space from row reduction

For a matrix A , the column space and null space are subspaces. Row-reducing A identifies both: the null space is read from the parametric solution of $AX = 0$, and the columns of A in the pivot positions span the column space.

6.3 Sum, intersection, and complementary subspaces

Definition 7 For subspaces F, G of E :

- the **intersection** $F \cap G = \{u : u \in F \text{ and } u \in G\}$ is a subspace;
- the **sum** $F + G = \{u + v : u \in F, v \in G\}$ is a subspace, the smallest containing both.

The sum is **direct**, written $F \oplus G$, when $F \cap G = \{0\}$. If moreover $F \oplus G = E$, then F and G are **complementary** subspaces of E : every $u \in E$ writes uniquely as $u = u_F + u_G$ with $u_F \in F, u_G \in G$.

7 Linear independence, bases, dimension

Definition 8 Vectors $v_1, \dots, v_k$ are **linearly independent** if the only scalars with

$$\lambda_1 v_1 + \dots + \lambda_k v_k = 0$$

are $\lambda_1 = \dots = \lambda_k = 0$. Otherwise they are **linearly dependent**.

They form a **generating family** of E if $\text{span}(v_1, \dots, v_k) = E$.

A **basis** of E is a family that is both linearly independent and generating.

Theorem 9 All bases of a vector space E have the same number of vectors. This common number is the **dimension** of E , written $\dim E$.

For a matrix in row echelon form, the **pivot columns** of the original matrix form a basis of its column space, and the **rank** of the matrix — the number of pivots — equals the dimension of the column space.

Proposition 10 In a space of dimension n :

- any linearly independent family has at most n vectors;
- any generating family has at least n vectors;
- any family of exactly n vectors that is independent **or** generating is a basis.

A basis can be **extracted** from any generating family by discarding vectors that are linear combinations of the others.

8 The Four Dimensions Theorem

Theorem 11 (Four Dimensions) For finite-dimensional subspaces F, G of a vector space,

$$\dim(F + G) = \dim F + \dim G - \dim(F \cap G).$$

In particular $F \oplus G = E$ (direct sum filling E) requires both $F \cap G = \{0\}$ and $\dim F + \dim G = \dim E$.

9 Linear maps

Definition 12 A map $f : E \rightarrow F$ between vector spaces is **linear** if for all $u, v \in E$ and $\lambda \in \mathbb{R}$:

$$f(u + v) = f(u) + f(v), \quad f(\lambda u) = \lambda f(u).$$

Definition 13 For a linear map $f : E \rightarrow F$:

- the **kernel** is $\ker f = \{u \in E : f(u) = 0\}$, a subspace of E ;
- the **image** is $\operatorname{im} f = \{f(u) : u \in E\}$, a subspace of F ;
- the **rank** is $\operatorname{rk} f = \dim(\operatorname{im} f)$.

Theorem 14 (Rank theorem) For a linear map $f : E \rightarrow F$ with E finite-dimensional,

$$\dim E = \dim(\ker f) + \operatorname{rk} f.$$

9.1 Injectivity, surjectivity, bijectivity

Proposition 15 Let $f : E \rightarrow F$ be linear, with E, F finite-dimensional.

- f is **injective** if and only if $\ker f = \{0\}$.
- f is **surjective** if and only if $\operatorname{rk} f = \dim F$.
- f is **bijective** if and only if both hold; this requires $\dim E = \dim F$.

10 Matrix representation and change of basis

Definition 16 Let $\mathcal{B} = (e_1, \dots, e_n)$ be a basis of E . Every $u \in E$ writes uniquely as $u = x_1 e_1 + \dots + x_n e_n$; the column $(x_1, \dots, x_n)^T$ is the **coordinate vector** of u in $\mathcal{B}$.

Definition 17 Let $f : E \rightarrow F$ be linear, $\mathcal{B} = (e_1, \dots, e_n)$ a basis of E and $\mathcal{C}$ a basis of F . The **matrix of f** relative to $\mathcal{B}$ and $\mathcal{C}$ has, as its j -th column, the coordinate vector of $f(e_j)$ in $\mathcal{C}$.

If X is the coordinate vector of u in $\mathcal{B}$, then AX is the coordinate vector of $f(u)$ in $\mathcal{C}$.

10.1 Change of basis

Let $\mathcal{B}$ and $\mathcal{B}'$ be two bases of E . The **change-of-basis matrix** P from $\mathcal{B}$ to $\mathcal{B}'$ has, as its columns, the coordinate vectors of the vectors of $\mathcal{B}'$ expressed in $\mathcal{B}$. If X and X' are the coordinate vectors of the same u in $\mathcal{B}$ and $\mathcal{B}'$ respectively, then

$$X = PX'.$$

Change of basis is introduced here and developed further in Linear Algebra II.